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Review Article

Glutamine Supplementation: A Possible Strategy to Help Mitigate Health Risks of Heat-Related Illness

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A rise in body temperature caused by physical work, including exercise, in a hot climate can lead to heat-related illnesses such as exertional heat exhaustion and stroke. Individuals who work physically demanding occupations in hot environments are at heightened risk of heat injury. The mechanisms that contribute to heat illness resulting from physical work in the heat are complex and include dehydration, tissue ischemia and damage, oxidative stress, and inflammatory events. Therefore, it is important to develop strategies that address these mechanistic underpinnings to prevent exacerbation to heat illness. Glutamine is an amino acid that has been considered conditionally essential during situations of biological stress (e.g., tissue burn, exercise, sepsis) due to high rates of tissue consumption. Evidence suggests that glutamine may serve as an important nutrient during heat stress and when combined with other preventative measures (e.g., cooling techniques, work/rest ratios, clothing) may help to mitigate heat illness among individuals working in extreme climates. The aim of this review is to examine the current literature on the role of glutamine during heat stress.

Keywords: exercise; glutamine; heat; illness; occupation

1. Introduction

Physical work in a hot environment can cause an increase in core temperature (hyperthermia), excessive fatigue, dehydration, and impaired cognitive performance [1]. These factors contribute to exertional heat illnesses which include exertional heat exhaustion (EHE); exertional heat injuries (EHI), such as gastrointestinal (GI), kidney, liver, or muscle damage as a result of hyperthermia; and more severe exertional heat stroke (EHS), i.e., a life-threatening condition characterized by hyperthermia along with central nervous system failure [2]. Exertional heat illnesses develop from external heat exposure combined with internal heat production (i.e., heavy physical work), which cause an increase in core temperature that is managed via the cardiovascular system's ability to increase blood flow to the skin surface for heat dissipation [1]. Myocardial strain is exacerbated by declining circulating volume (via sweating), and competition between exercising muscle and skin for blood delivery [3]. The re-direction of blood flow to the skin results in a reduced delivery to other tissue beds, resulting in relative hypovolemia. Various tissues such as such as brain, gut, and liver are often affected [4–7]. Prolonged sweat loss then causes a situation of absolute hypovolemia, and if severe, ischemia can arise causing tissue damage and production of circulating free radicals, inflammatory agents (e.g., cytokines), and endotoxins [4, 8–10]. The management of core temperature during heat stress (HS) is paramount to mitigating related illness and injury; however, the degree to which these downstream events occur (e.g., tissue ischemia, inflammation, and circulating molecules) during mild exertional hyperthermic events is not well known [11]. Evidence suggests that HS during prolonged exercise can cause both intestinal wall damage and further upregulate inflammatory markers [11]. In addition, the knowledge of daily work in the heat (e.g., construction laborers, agriculture workers) on markers of heat illness is also incomplete among humans. e.g., a high prevalence of impaired renal

function was reported among sugarcane cutters, which was attributed to the physiological impact of daily work in a hot and humid environment [12].

The complexities of prolonged physical work in the heat present a challenge for developing and implementing prevention strategies. Novel intra-event cooling strategies such as water- and air-cooled vests, cold towels, menthol shirts, and ice packs have been shown to delay hyperthermia but a rise in core temperature eventually occurs [13, 14]. Hydration strategies have also been explored to help mitigate cardiovascular strain and tissue damage [15, 16]. Markedly less is known about the potential benefit of dietary supplements for mitigating heat illnesses [17]. Various antioxidants, dietary nitrates, amino acids, probiotics, and creatine have been tested among humans and animals on various physiological outcomes of HS [17]. The aims of these studies were to identify a supplement that can match the mechanistic underpinnings of heat illnesses (e.g., hydration, inflammation, oxidative stress), and when combined with cooling strategies, may provide a comprehensive approach for prevention.

Evidence suggests that glutamine is an ideal supplement to serve this role due to versatility of the amino acid in supporting tissue protection, immune function, and hydration [18, 19]. The aim of this review is to provide a critical exploration of glutamine's role in protecting humans against exertional heat illness. Included is an examination of mechanisms related to tissue damage, inflammation, and hydration, along with cellular and oxidative stress. To justify the importance of novel strategies to combat heat illnesses, a discussion regarding the prevalence and potential economic and mortality impacts will be highlighted as well. Throughout the review, the adopted term HS is used to identify physical work, including exercise, that leads to a rise in core temperature.

2. Prevalence of Exertional Heat Illnesses

Individuals working in physically demanding professions such as construction, agriculture, firefighting (both structure and wildland), police, warfighters, manufacturing, and warehousing are at heightened risk for EHI. Workers are four times more likely to experience heat injury (e.g., dehydration, palor, nausea, vomiting, confusion) after a single shift under hot conditions when compared to workers in ambient temperatures, and 15% of individuals who consistently work in the heat are susceptible to heat-related injuries [20]. Most vulnerable are individuals from lower-income groups, casual workers, construction laborers, those with preexisting health concerns, aging workers, and those with a history of heat illness [21–23]. The global estimates suggest that a fatal case of occupational HS is expected to occur every 14–24 min globally by 2030 [24]. The workforce productivity loss due to heat in the United States is projected to be 0.21%, which is equivalent to nearly 400,000 full-time jobs with an economic impact exceeding $2.4 trillion [21, 25].

Athletes and military service members are also at heightened risk for heat illness. Among competitive athletes, EHS, which is the most severe exertional heat illness, is a leading cause of sudden death [26, 27]. American football has historically had the highest rates of heat-related fatalities [28]. Determining the incidence of less severe EHIs such as heat injury or heat exhaustion among athletes has been much more difficult. Between 2005 and 2009, The U.S. Centers for Disease Control and Prevention (CDC) reported a weighted yearly average of 9237 cases of EHIs among high school athletes. The NCAA Injury Surveillance Program (166 institutions and 2048 team-seasons) received reports of only 232 exertional heat illnesses during sanctioned practices and competitions among collegiate athletes between 2009 and 2014 [29]. It's unknown if a lack of reporting exists, and cases of exertional heat illnesses during nonsanctioned practices are undetermined. Collegiate sports that put athletes at greatest risk were American football and soccer (both men and women). Rates among recreational athletes was reported to be nearly 6000 cases per year between 2001 and 2006, but current data do not exist [30]. There is a clear need to improve recording and reporting of nonfatal exertional heat illness occurrence among athletes at all levels; however, the immense challenge of such an endeavor is extraordinary.

Recent reports among U.S. Armed Service members indicated that in 2022, EHS and heat exhaustion occurred at rates of 32.1 and 147.7 per 100,000 person-years, respectively, with a total number of 12,228 cases between 2018 and 2022 [31]. At greatest risk were males and individuals between 20 and 29 years of age. Interestingly, only 3.1% of reported heat illnesses among military personnel occurred outside of the United Sates between 2018 and 2022. The U.S. military has been forthright in documenting and reporting cases since 2001; comparatively, detailed data from military forces across the globe is rarely published [32].

Organizations such as the National Institute for Occupational Safety and Health (NIOSH) and National Athletic Trainer's Association (NATA) have published recommendations for preventing and treating exertional heat illnesses [33, 34]. These have included adjusting work/exercise and rest cycles, reducing physical effort, enhancing heat tolerance via acclimation, worker and athlete training and recognition of symptoms, access to cool potable drinking water, and medical screening. A review of OSHA heat enforcement demonstrated that over 80% of employers did not rely on national standard approaches for heat injury mitigation [35]. Moreover, it was recently demonstrated that compliance with heat prevention guidelines was insufficient for reducing risk for heat illness among California farm workers [36]. This may indicate that further heat mitigation strategies are necessary.

3. Physiological Response to Heat Strain

Heat strain occurs when an imbalance of heat production and dissipation results in a rise in core temperature, which can be partially mitigated via the sweat response. Unless fluids are actively consumed to match fluid loss, a significant amount of water depletion (i.e., absolute hypovolemia) will occur, and can exacerbate many of the consequences of HS [37]. Dehydration paired with hyperthermia from HS leads to cardiac strain and increased blood viscosity, ultimately

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resulting in decreased systemic blood flow. A consequential response is relative hypovolemia from the shunting of blood away from splanchnic circulation to assist with core temperature control causing a milieu of reactive oxygen species (ROS) [38]. Oxidative stress within the intestinal epithelial membranes degrades the integrity of tight junction (TJ) proteins and facilitates endotoxin leakage into the portal circulation, and further release of inflammatory mediators (i.e., cytokines and chemokines) via circulating leukocytes (fig) [39].

The central role of GI tract in EHIs was first hypothesized by Snipe et al., and further human experiments demonstrated the GI permeability–endotoxin–inflammation pathway after exercise in the heat [11, 40, 41]. Studies have elucidated that heat can induce alterations in the expression and localization of TJ proteins, such as occludin, claudins, and zonula occludens, leading to increased intestinal permeability [42, 43]. Compromised integrity of TJs permits paracellular passage of luminal antigens, toxins, and pathogens into the bloodstream (see Figure 1). GI insult is exacerbated during higher intensity bouts of exercise, and damage is also associated with the rise in the core temperature [44, 45]. Protecting the intestine during prolonged and intense HS is considered a central preventative measure against exertional heat illnesses [46]. Maintenance of upstream mechanisms such as hydration and preservation of circulating blood volume is another important preventative measure. Considerations must also be made regarding the inflammatory response that often follows GI insult and endotoxin leakage. Therefore, a treatment or supplement that helps preserve hydration, offers GI protection, and mediates the immune response would be ideal in the defense against exertional heat illnesses and specifically EHI. The following sections provide further detail into each mechanism and the role of glutamine.

4. Glutamine's Role in the Body During Physical Stress

Glutamine is the most abundant amino acid in the body and is central for supporting metabolic intermediary metabolism, ammonia transport, and pH balance [47–49]. Isotopic techniques have estimated that endogenous production is 40–80 g/day, and approximately 70–80 g are distributed in the whole body (dependent upon body composition) making up ~20% of the circulating free amino acid pool [50, 51]. It is nutritionally classified as a nonessential amino acid; however, under conditions of trauma (e.g., severe burn, surgery), exhaustive physical exercise, and sepsis, tissue uptake and utilization of glutamine increases substantially [52, 53]. The rise in glutamine use during stress and catabolic conditions is met via intramuscular free glutamine and plasma levels, which both decline substantially under conditions of sepsis and physical trauma [54, 55]. Tissue uptake of glutamine during periods of stress supports metabolism, cellular defenses, molecular signaling, and glutathione (GSH) synthesis [51]. For example, a 50% decrease in postoperative plasma glutamine has been reported among surgical patients [56]. The intestinal tract, immune cells, and liver are believed to be major consumers of glutamine during periods of high stress [57, 58]. The interorgan and tissue flux support cellular integrity in the intestine; immune system proliferation and energy metabolism; and regulation of antioxidant defenses and nitrogen metabolism in the liver [59, 60]. The high rates of glutamine consumption during stress have led to the amino acid being considered conditionally essential [19].

Circulating glutamine levels appear to be stable during long duration (2 h) exercise but are shown to decrease by 16%–20% in recovery [61, 62]. Postexercise decline may be duration dependent with prolonged endurance bouts causing large drops in glutamine [63, 64]. For example, 140 min of cycling until exhaustion caused a 19% (682 to 552 μmol^{-1}) decrease in plasma glutamine that was detected after 2 h of recovery and returned to baseline by 7 h of recovery [65]. Limited evidence also suggests that high-intensity interval exercise lasting more than 30 min lowers recovery plasma glutamine [66]. The liver is a major consumer of plasma glutamine during periods of prolonged stress (e.g., physical exercise) where hepatic glutamine uptake has been shown to increase sixfold during exercise in dogs [67]. A large amount of glutamine uptake during exercise was used for nitrogen metabolism and urea production [68–70]. Hepatic glutamine is also a likely predominant gluconeogenic amino acid for blood glucose maintenance during prolonged exercise [71]. Glutamine use in the liver during exercise may further contribute to combatting oxidative stress via GSH synthesis through induction of the erythroid 2-related factor 2 (Nrf2) pathway [72, 73].

The gut and lymphocytes are also avid consumers of glutamine during physical stress such as exercise. For example, Krishna et al. demonstrated a two- to threefold increase in intestinal glutamine extraction during exercise in dogs [67]. Glutamine's actions in intestinal epithelial cells during stress are to support enterocyte metabolism, dietary nitrogen and carbon processing, cell proliferation, and gut barrier maintenance [74–77]. The GI tract is considered the principal consumer of glutamine with evidence suggesting that > 50% of enterally administered glutamine is metabolized within the intestinal mucosa [78]. Glutamine may be a more efficient energy fuel for enterocytes as evidence shows a yield of 9 ATP/1 mol glutamine vs. 8 ATP/1 mol glucose [79]. The nonaerobic glycolytic pathway supplies the bulk of the ATP for enterocytes, and glutamine utilization may spare both endogenous and exogenous glucose substrates to support other tissues [75, 79–81].

Glutamine is an important biosynthetic and metabolic nutrient for both T and B lymphocytes, macrophages, and peripheral blood mononuclear cells during prolonged exercise and other situations of physical stress [82]. These cells all demonstrate high rates of glutaminase activity, and accelerated rates of glutamine utilization have been reported to be greater than glucose uptake [82, 83]. Through the process of glutaminolysis, glutamine is converted to glutamate, asparate, and alanine to support cellular metabolism and is essential for both T and B cell regulatory function [84, 85]. The oxidation of glutamine for energy generation

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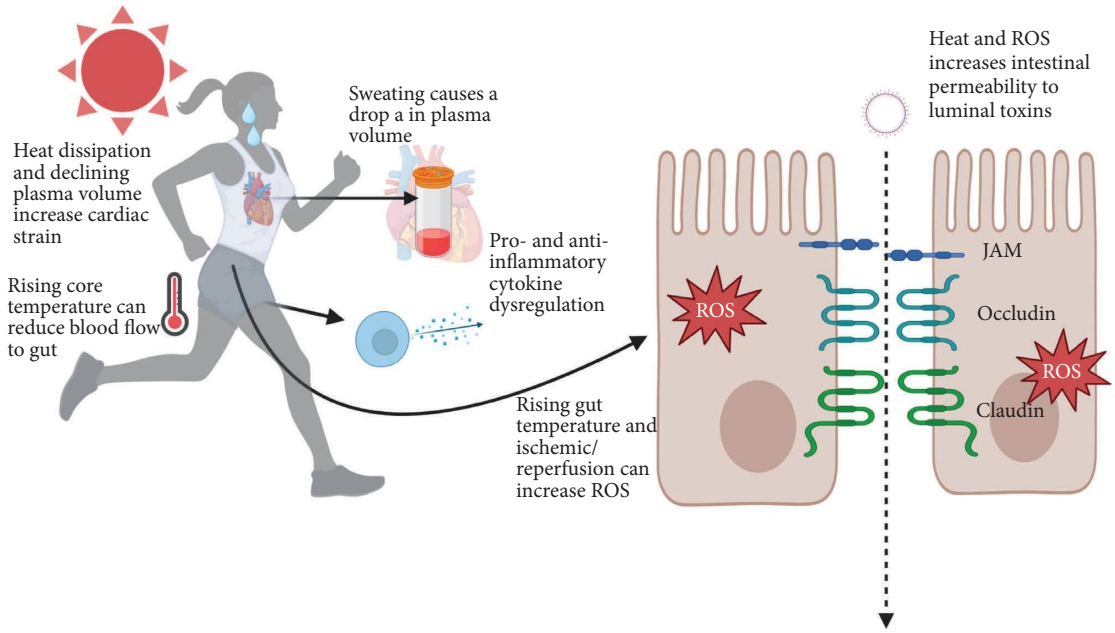

Figure 1: The central role of the gastrointestinal damage during physical work in the heat. Created with BioRender.com.

within lymphocytes and macrophages is incomplete which indicates that the amino acid may be important for other functional roles within immune cells. Specifically, glutamine can be used to support the synthesis of purines and pyridines, along with induction of protein transcription factors that regulate cellular proliferation (c-jun N-terminal kinases, JNK, and activator protein-1, AP-1) and cellular repair (heat shock proteins, Hsps) [47, 86]. Glutamine and glutamate can also be used to generate essential compounds in neutrophils for defense against ROS [87]. Glutamine may also regulate both pro- and anti-inflammatory cytokine signaling in various immune cells under conditions of illness and physical stress such as extreme exercise. Glutamine consumption by leukocytes helps to explain skeletal muscle glutamine depletion during situations of bodily stress. Glutamine is also used to synthesize antioxidant defenses in red blood cells and has emerged as an important nutrient to prevent complications of sickle cell disease [88].

Shifts in glutamine tissue transport and utilization during HS is not well understood in humans. Sixty minutes of treadmill running at 70% intensity in a heated environment (30°C) caused a small nonsignificant decrease in circulating glutamine levels at immediately after, 2 h, and 4 h postexercise time points [9, 89, 90]. Levels decreased significantly after 87 min of exercise in a markedly hotter environment (38°C) that included various activities such as treadmill walking, resistance exercises, and box stepping [91]. It is clear that an increase in plasma glutamine levels through oral supplementation results in a substantial decline in glutamine at post-HS time points. For example, in numerous studies, our group and others have reported a significant increase in glutamine levels after supplementation followed by a decline after HS that can be further rescued with continued glutamine ingestion [9, 89–92]. The sharp decrease after HS may indicate glutamine tissue uptake and utilization in various biochemical reactions. The following sections highlight evidence of glutamine's role in these systems during HS and other related stressors. Figure 2 provides a summary.

4.1. Intestinal Damage/Dysfunction During Heat Stress. The role of glutamine in protecting the gut during heat exposure is best identified by examining glutamine supplementation trials. Animal and intestinal cell culture studies have shown protective effects of glutamine supplementation during heat exposure interventions [93–95]. Soares et al. reported improved intestinal barrier function and reduced bacterial translocation in mice that were fed a glutamine-supplemented diet versus isoenergetic and isonitrogenous placebo chow for 7 days prior to passive HS [96]. Lower levels of plasma endotoxin and increased survival following classic heat stroke were demonstrated in rats after 5 days of glutamine ingestion compared to control [97]. Heightened resistance to injury and apoptosis has been further shown in glutamine supplemented intestinal cell cultures after lethal heat insult [98, 99].

Our group has reported improved intestinal barrier function and lower damage after oral glutamine ingestion among humans exercising in the heat. The rise in intestinal permeability was abolished after 60 min of high-intensity treadmill running when high-dose (0.90 g/kg/day) glutamine was ingested for 7 days prior (chronic supplementation) or immediately before (1 single dose) [9, 89]. Pugh et al. tested the effect of glutamine dose (0.00, 0.25, 0.50, 0.90 g/kg/day) on intestinal permeability among humans after 1 h of treadmill exercise in the heat, and found that glutamine attenuated permeability in a dose-dependent fashion [90]. Acute glutamine ingestion (dosed at 0.30 g/kg/day) prior to 87 min of HS also led to reduced intestinal damage,

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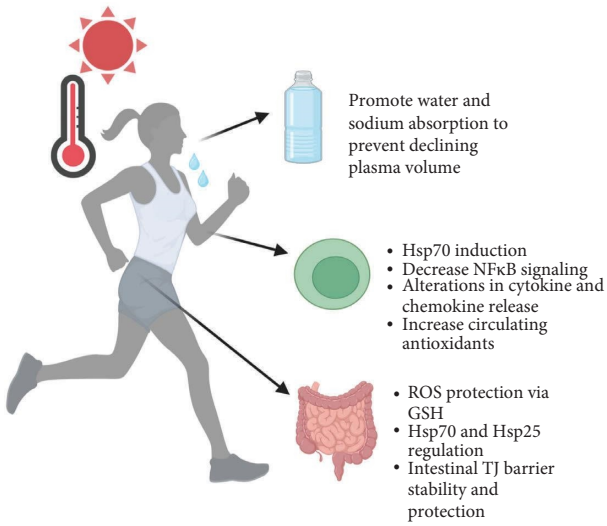

Figure 2: The role of glutamine during physical work in the heat. Evidence suggests that glutamine ingestion can promote hydration, regulate cellular defenses through Hsp70 induction and regulation of inflammatory mediators, along with stabilizing the intestinal epithelial wall. Created with BioRender.com.

identified by plasma fatty acid-binding protein (IFABP), that was observed 24 h after HS [91, 100]. Circulating IFABP levels after a 20-km time trial in the heat were lower among cyclists who ingested a glutamine bolus (0.90 g/kg) versus a placebo prior to exercise. Conversely, acute low-dose oral glutamine (0.30 g/kg/d) ingestion had no effect on intestinal permeability and IFABP changes after exhaustive running (~22 min) in heat and after 80 min of treadmill walking in the heat [101, 102]. The lack of consistent findings may be explained by the measured time points as intestinal permeability was captured 1 h postexercise compared to 5 h postexercise in previous studies. Plasma IFABP was also collected at the postexercise time point only, versus 2-, 4-, and 24-h postexercise in similar trials [91]. Another explanation is that the HS was not severe enough to cause intestinal disruption [45]. However, the null findings may also suggest that a higher dose of glutamine is required before HS and that benefits favor chronic supplementation over a single dose.

Several mechanisms explain the protective benefits of glutamine on intestinal barrier stability and function during HS. First, glutamine has been shown to activate heat shock factor-1 (Hsf-1), the transcription factor for inducible heat shock proteins or Hsps (namely Hsp70/72 and Hsp25) [99, 103]. The heat shock response is a cytoprotective event that promotes cell survival through protein re-folding and trafficking combined with modulation of inflammatory events via regulation of the NFκB pathway [104, 105]. Mild stress causes an increase in Hsp70 and Hsp25 levels in intestinal epithelial cells designed to maintain cellular structure and function. Upon prolonged or more severe stress, inflammatory mediators such as TNF-α and IFN-γ increase and downregulate Hsp70 and 25 [106, 107]. Evidence suggests that glutamine is necessary for Hsp70 and Hsp25 stabilization in intestinal cells during HS where glutamine inhibition diminishes Hsp induction and cellular protection [99, 103]. Glutamine promotes Hsf-1, Hsp70, and Hsp25 gene induction through the hexosamine biosynthetic pathway (HBP), which splits from glycolysis at the fructose-6-phosphate step [108, 109]. HBP metabolism of glutamine leads to *O*-glycosylation (via *O*-linked *N*-acetylglucosamine or GlcNAc), nuclear translocation, and transcriptional regulation of Hsf-1 and Sp-1 and thus Hsp70 and Hsp25 synthesis [108, 110]. The O-GlcNAc transferase (OGT) enzyme catalyzes GlcNAcylation of proteins in mammals, and OGT silencing prevented glutamine linked *O*-glycosylation and activation of Hsf-1 and Hsp70 in heat stressed mouse fibroblasts [108, 111].

Heat shock proteins protect the gut during HS by stabilizing intestinal TJ proteins. In a Caco-2 cell model, Dokladny, Moseley, and Ma demonstrated that Hsp70 was responsible for stabilization of occludin during physiologically relevant thermal insult [112]. Hsp70 may exert an effect on occludin by physical binding and stabilization, which has been shown to occur during heat exposure [113]. Hsf-1 and Hsp70 induction was also linked to zona occludens and claudin-1 mRNA expression in rat jejunum after HS [114]. Occludin, claudins, and zona occludens (ZO-1, ZO-2, and ZO-3) are key proteins that regulate the TJ barrier and paracellular transport in the gut. Another mechanism for Hsp protection in the gut is by regulating the state of intestinal epithelial actomyosin contraction. The TJ is influenced by the contractile state of the actin-myosin cytoskeleton within adjacent intestinal epithelial cells [115]. High temperature environments can cause phosphorylation of myosin light chain kinases (MLCK) via protein kinase C (PKC) with increased permeability as a result [116]. Pretreatment of T84 cells (human colon-derived crypt-like cell line) with Hsp70 effectively inhibited PKC phosphorylation of MLCK during HS [116]. In summary, Hsps prevent intestinal permeability through physical stabilization and protection of TJ proteins along with regulating contraction of the actomyosin chain. Jejunal enterocytes supplemented with glutamine had increased transepithelial electrical resistance (TEER) in accordance with occludin, claudins, and ZOs expression, which may be linked to glutamine regulation of the Hsp family [117].

Another mechanism that may explain glutamine protection in the gut is through the control of mitogen-activated protein kinases (MAPK) and phosphatidylinositol 3-kinase (PI3K)/AKT pathways. Key MAPKs are extracellular signal-related kinase (ERK), JNK, and p38 MAPK [118]. ERK signals cell proliferation, DNA synthesis, and promotes anti-apoptosis while JNK and p38 are stress related and lead to cell death in many cell lines, including the intestinal epithelial cells [119]. Larson et al. demonstrated that glutamine mediates cell survival through ERK phosphorylation and activation [119].

A third mechanism for glutamine intestinal benefits is through the control of inflammatory mediators in the gut during various forms of physical stress. Hou et al. reported decreased colonic expression of the NFκB inflammatory pathway in mice supplemented with glutamine prior to chemical insult [120]. In intestinal epithelial cells, NFκB is

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a transcription factor that controls inflammation, innate immunity, and tissue damage [121]. NFκB is often considered a double-edge sword where on one hand, gene silencing exacerbates tissue damage and inflammation and on the other hand, overexpression of NFκB does the same [85, 122]. This is evident in many chronic inflammatory disorders such as inflammatory bowel disease (IBD), rheumatoid arthritis, and coronary artery disease [123]. Unstimulated NFκB is physically bound in the cytosol by the IκB kinase (IKK) complex consisting of IκBα and IκBβ subunits [124, 125]. Activation results in phosphorylation and degradation of IκBα, followed by nuclear translocation of NFκB into the nucleus for the transcription of numerous genes, including cytokines (e.g., IL-1, IL-6, IL-8, TNF), chemokines (e.g., MCP-1), adhesion molecules (e.g., ICAM-1), and immune cell activity [126, 127]. Oxidative, viral, and microbial cytokine stressors along with protein signaling such as PKC and MAPK are known activators of NFκB [125, 128]. While unclear in human studies, thermal insult has been shown to induce intestinal NFκB activation in animal studies [129]. For example, Xia et al. reported heightened NFκB activity in intestinal tissue of heat stroke mice that were associated with elevations in circulating TNF-α, IL-6, and IL-1β [130]. Similar findings were presented in the ileum mucosa of pigs after 7 days of chronic HS [131]. Glutamine may regulate NFκB activation by decreasing MAPK and PKC signaling, which was observed in colonic tissue of individuals with Crohns supplemented with glutamine [132, 133]. Supplementation also prevented IκBα degradation, NFκB induction, and downstream cytokine synthesis during inflammatory insult in human intestinal cells, indicating another site of glutamine control [134]. Glutamine upregulation of Hsp70 has been shown to physically bind to IκBα and prevent ubiquitination, which likely explains the mechanism of glutamine stabilization of IκBα [134, 135].

Glutamine also controls inflammation in the gut by modulating free radicals that can be generated from the ischemic-reperfusion cycling that occurs during hyperthermia [136]. ROS production generated from oxidative stress has been proposed as a heat illness-related pathology [136]. In this model, blood flow is re-directed away from the gut to support thermoregulation, resulting in cellular hypoxia, production of ROS, and tissue damage [136]. Daily HS was shown to increase production of ROS and decrease antioxidant activity in rat intestinal epithelial cells but treatment with glutamine protected the cells against both heat and oxidative damage [98, 137]. Glutamine serves as a precursor for GSH synthesis, which is a central cellular antioxidant [52]. Intracellular levels of GSH were increased in the intestine, liver, lung, and muscle after glutamine ingestion [138]. In the gut specifically, Cao et al. demonstrated that glutamine metabolism provided glutamate to synthesize GSH in rat intestinal tissue, and glutamine fed animals had a threefold higher expression of GSH [139]. Glutamine also preserved GSH levels during intestinal ischemic reperfusion insult in animals [140]. The ratio of reduced (GSH) and oxidized (GSSG) intracellular GSH determines the redox potential, and ROS production and scavenging by GSH during ischemic events lowers the ratio. Intestinal antioxidant defenses are dependent upon maintaining a high cellular concentration of GSH in proportion to GSSG, and glutamine serves as a precursor to support maintenance of this ratio [141].

In summary, glutamine protects the gut during the HS by stabilizing the TJ barrier via preservation of inducible heat shock proteins (Hsp70 and Hsp25), and through control of the intestinal epithelial cell cytoskeleton. Glutamine modulates the inflammatory response that accompanies prolonged HS via control of the NFκB and GSH pathways.

4.2. Immune Support During Heat Stress. The immune response during classic heatstroke (i.e., passive heat exposure) has been characterized by elevated levels of inflammatory cytokines that have been monitored at the time of collapse or upon cooling. Bouchama et al. monitored pyrogenic cytokine levels among 28 classic heatstroke patients and reported elevated IL-6, IL-1β, and INF-γ in all patients, and IL-6 concentrations were associated with severity of heat illness [142]. Similarly, elevated levels of IL-6 were detected upon hospital admission among EHS patients [143]. A correlation between inflammatory cytokine activity and core temperature changes during heatstroke has been challenging due to the complexity between pro- and anti-inflammatory signaling. For example, Hashim et al. observed elevations in anti-inflammatory cytokines, IL-1ra and TNFs r-II, among exertional heatstroke patients, which inhibit IL-1 (β and α) and TNF-α, respectively [143–145]. Higher levels of TNFsr-II were associated with mortality. Moreover, classic heatstroke in a baboon model was characterized by a rise in anti-inflammatory cytokine, IL-10 and chemokine, IL-8 along with IL-1ra and TNFs r-II that were observed early (~1 h) into the heat insult but mortality followed [146]. This reveals that an exaggerated anti-inflammatory cytokine response may worsen heatstroke outcomes, which is further demonstrated by IL-6 and TNF-α knockout mice showing higher classic heat stroke-related mortality [147]. Together, these findings indicate that severe heat illness causes dysregulation of anti-inflammatory modulators and likely play a role in the pathology of both exertional and classic forms of heatstroke. In the only known study examining glutamine supplementation on cytokine release after classic heatstroke, it was recently reported that 7 days of glutamine supplementation lowered IL-8 and mitigated the rise in IL-6 levels among rats after heatstroke compared to placebo control indicating a modulation of inflammatory cytokines, but the changes in anti-inflammatory cytokines were not reported [148].

The inflammatory response to nonlife threatening HS and the impact on systemic function is not well defined. Snipe et al. reported higher peak IL-6, IL-10, and IL-1ra values among healthy adults after 120 min of running in the heat compared to a nonheat running control [11]. The research group also explored anti-inflammatory to pro-inflammatory ratios measured as IL1β:IL-10 and TNF-α:IL-10 ratios and reported lower ratios throughout the HS trial [11]. These findings align with heightened anti-

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inflammatory responses during EHS noted above. Interestingly, Hailes et al. observed heightened basal levels of several anti-inflammatory agents such as IL-10, IL-1ra, and TNFsr-II after four consecutive days of HS, and suggested a modulated inflammatory profile that may alter the immune response to additional heat exposures [149]. Conversely, pro-inflammatory agents such as TNF-α, IL-1β, IL-12p40, and MCP-1 have been shown to increase after exercise in the heat [41, 149, 150]. Circulating cortisol and LPS are possible mediators of the inflammatory response as both increase after HS compared to exercise control [11]. While the pro- and anti-inflammatory responses to HS are complex, it's clear that prolonged heat exposure can cause dysregulation of this system, and put one at risk for exertional heat illnesses, especially among individuals who are unacclimated [149, 150].

Like glutamine's role in intestinal tissue during HS, the effect of the amino acid on immune function during HS is multifactorial. Neutrophils, macrophages, and lymphocytes are all large consumers of glutamine, and utilization rates rise during cellular stress to support metabolic activity and promote protein signaling pathways that influence cellular defenses along with cell proliferation and cytokine production [18, 47]. Zheng, Chen, and Zhou monitored all circulating immune cell types after exhaustive HS among untrained males who consumed 0.6 g/kg of glutamine 30 min prior to exercise [151]. Only T cell lymphocyte mobilization into the peripheral blood was higher in the glutamine trial compared to HS alone [151]. Namely, the responsiveness of CD3+ and CD3+CD8+ T lymphocytes was observed, which regulates the activities of T helper and cytotoxic T naïve cells. Cell model experiments have demonstrated that glutamine is essential for T lymphocyte proliferation [152]. This may indicate heightened immune surveillance to invading pathogens. In animal models, consecutive days of heat exposure led to a depression in CD3+ and CD8+ T cells in broiler chickens [153]. T lymphocytes are capable of releasing pro-inflammatory cytokines such as IL-6, IL-2, IL-1, and TNF-α; however, cytokine levels were unchanged immediately after the glutamine supplemented HS trial and explained by the low intensity (40%) and short duration (<40 min) of the trial. High-dose glutamine (0.90 g/kg) ingested prior to 60 min of high intensity HS (70%) led to lower TNF-α levels compared to placebo trial that was detected at 4 h into recovery [89]. Unfortunately, little is known about the cytokine and chemokine response to HS after glutamine supplementation. However, these findings may help to better focus researchers on T cell function and the role of glutamine [151].

Glutamine regulation of heat shock response also plays a role in circulating immune cells during HS. Both acute and 7 days of high dose (0.90 g/kg) prior to exercise in the heat increased Hsp70 expression in PBMCs observed at 2- and 4-h postexercise [89, 154]. Glutamine dosed at 0.30 g/kg and administered before and after consecutive days of prolonged HS increased PBMC Hsp70 levels after the second day [91]. In a similar trial, 0.15 g/kg of glutamine (dosed before and after HS) increased Hsp70 expression 4 h after HS on the first day compared to a placebo supplement [92]. As reviewed in intestinal epithelial cells, glutamine induction of Hsp70 may insert regulatory control of the NFκB pathway through IκBα stabilization and reduction in inflammatory cytokine release from PBMCs [89, 91, 92, 155]. However, the effect of glutamine combined with HS on Hsp70 activation, IκBα preservation, and NFκB inhibition in PBMCs have all been associations, and gene silencing models do not existing. Therefore, it's unknown if these findings are cause and effect or mediated through other regulatory pathways. However, glutamine's attenuation of NFκB was shown to be dependent on Hsp70 activation in lung tissue from an animal sepsis model [97].

Glutamine may also play a role in antioxidant maintenance during HS, but limited human studies exist. Tissues other than the intestines are vulnerable to oxidative stress during prolonged heat exposure. Evidence has suggested that myocytes, skeletal muscle tissue, neurons, erythrocytes, and macrophages produce ROS during heat exposure [156]. Mitochondria are the primary source as ROS are the byproducts of partially reduced oxygen that occurs during high rates of oxidative phosphorylation that can occur during heat exposure [156–158]. Key antioxidant systems that mitigate mitochondrial ROS include GSH, catalase, and superoxide dismutase [156]. Glutamine supplementation in broiler chickens exposed to HS resulted in higher levels of all three mentioned antioxidants regardless of tissue type [159]. In the only known human study, low-dose glutamine supplemented at 0.15 g/kg increased plasma total antioxidant capacity (TAC) 4 h after 78 min of HS [92]. The measure of TAC is used as an indirect measurement of systemic antioxidant levels and monitors the effects of antioxidant supplementation [160]. HS may exacerbate disorders linked to ROS, and chronic heat exposure may accelerate the negative impact of these molecules. Therefore, more work is needed to better understand glutamine's role in tissue antioxidant defenses during HS.

4.3. Hydration. Body water maintenance is pivotal to sustaining thermoregulation during long duration physical work in the heat. Dehydration by more than 2% can impair cognitive tasks such as attention and executive function which are essential in any work environment [161]. The impact of water loss on physical performance is not clear with declines in the range of 2%–4% linked to performance decrements in long duration activities [162]. Rehydration solutions and sports drinks often contain high amounts of sodium and glucose to promote water absorption and electrolyte replacement. American football athletes lost 2.5 g/h of sodium during a 4.5-h practice session (WBGT 25.9°C), and outdoor workers' (WBGT 29.3) loss ranged from 4.6 to 6 g over an 8- to 12-h work shift [163, 164]. These findings indicate that sodium replacement is important for individuals exposed to extreme physical and environmental challenges. Sodium ingested either within a hydration drink or in food increased fluid retention and restoration of plasma volume after a 2.5% body weight loss [165]. The addition of glucose promotes sodium absorption, glucose uptake, and water transport through the sodium-glucose co-

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transporter-1 (SGLT-1) channel [166, 167]. The linkage of glucose with obesity and diabetes has created some controversy regarding the value of the ingredient in beverage formulas. Evidence suggests that glutamine stimulates sodium and water absorption from the human jejunum, and may be considered as a substitute for glucose [168]. Coeffier et al. reported an increase in both sodium and water absorption after an infusion of a glutamine-enriched hydration mix among healthy adults [168]. Intestinal fluid and electrolyte uptake was maintained even in the face of induced secretory diarrhea [168]. Similarly, an L-alanyl-L-glutamine dipeptide infused drink was shown to increase sodium and water absorption along with improving physical and cognitive performance among males [169, 170]. The advantages of a glutamine beverage are effective hydration combined with intestinal and immune benefits that offers a multifaceted support against HS.

4.4. Fatigue. Fatigue is the failure of physical performance that results from both peripheral (i.e., skeletal muscle) and central (i.e., central nervous system) factors. Peripheral fatigue has been attributed to depletion of energy substrates, ammonia build-up, oxidative stress, and muscle proton build-up [171, 172]. Reduced neural motor drive along with changes in neurotransmitter such as increased serotonin and lowered dopamine are considered central fatigue mechanisms [171, 173]. Fatigue can also contribute to a decline in cognitive function that can occur during prolonged exercise or physical labor in a warm climate [174, 175]. Glutamine supplementation prior to prolonged exercise has improved markers of fatigue, but these findings have not been associated with improved physical performance [92, 176]. Regarding peripheral fatigue, glutamine serves as a major gluconeogenic amino acid in the liver and kidneys to support substrate metabolism during prolonged exercise [71]. Glutamine's role in energy metabolism prevents ammonia release, which occurs during amino acid oxidation in response to glycogen depletion during exercise [177]. Ammonia build-up during exercise is also attenuated because glutamine is the main nitrogen transporter and assists with ammonia removal. Bassini-Cameron et al. reported lower circulating ammonia in response to intense exercise among professional football players after chronic glutamine supplementation [177].

The impact of glutamine on mechanisms linked to central fatigue is less clear. It's unlikely that glutamine supplementation affects neural transmission since the blood brain barrier tightly regulates glutamine transport. Interestingly, evidence suggests that these transporters work by removing glutamine from the brain [19, 178]. Glutamine may work indirectly via the gut-brain axis, whereas the benefits of glutamine supplementation in the GI tract may contribute to changes in the central nervous system that improves cognitive function [179]. However, this is speculative as no studies have explored the effect of glutamine supplementation on cognition via the gut-brain axis. Other research also points toward glutamine supplementation having a positive effect on cognition, mood, and fatigue via the mechanisms highlighted in the previous sections, such as hydration, inflammation, and oxidative stress [180].

Several studies have shown that glutamine is an anti-fatigue amino acid, but limited data suggest improved physical performance as a result [176]. The mechanisms are likely linked to energy metabolism and ammonia regulation. Continued work is needed to better understand the influence of glutamine on fatigue in hot environments.

5. Glutamine Supplementation

The goal of glutamine supplementation is to increase circulating glutamine levels, which indirectly indicates a rise in intestinal glutamine uptake since > 50% of orally ingested glutamine is metabolized in the gut [78]. Glutamine dosed in the range of 0.15–0.90 g/kg/day increased plasma levels prior to HS [9, 90, 92]. Acute dosing at or above 0.90 has been linked to GI discomfort (e.g., bloating, diarrhea), which may indicate that an ideal dose-response exists [102]. Nava et al. administered 0.15 g/kg before and after HS (for a total of 0.30 g/day) for consecutive days (two HS trials) and observed a significant rise in pre-HS on day 1 (~23.80 mg/dL) and on day 2 (~28.97 mg/dL). Comparatively, a 0.90 g single bolus taken 2 h before HS increased (~23.82 mg/dL) to a similar level [91]. These results show that multiple small doses were effective in increasing glutamine levels compared to a single large dose. In addition, dosing before and after HS may prevent a decrease in circulating glutamine levels that is often observed during recovery from prolonged and intense exercise [65, 91].

Acute supplementation versus chronic glutamine ingestion on circulating glutamine levels in humans is not clear. Seven days of dosing at 0.90 g/kg split into three doses per day (0.30 g/kg/dose) increased glutamine levels prior to HS similar to a single 0.90 g/kg dose [9, 89]. Fourteen days of glutamine supplementation at 0.40 g/day had no effect on circulating glutamine levels among healthy adults who performed nine consecutive days of interval training. However, it was unclear when blood samples were collected (before or after training sessions or glutamine ingestion) as glutamine levels have shown to shift after heavy exercise in supplementation trials [89, 181]. These findings inform that physical activity patterns have a huge impact on circulating glutamine levels during supplementation cycles. In summary, acute glutamine ingestion may be ideal prior to a single bout of exercise or HS, and if an individual is chronically exposed, then continued glutamine ingestion may be helpful. The impact of chronic supplementation with and without daily physical work on markers of HS exposure is needed. In addition, it's unknown if daily glutamine ingestion improves one's ability to acclimate to environmental heat exposure. What we do know is that an acute dose can increase glutamine levels substantially and higher circulating levels may improve HS outcomes.

Based on the thorough review of the glutamine and HS literature, we suggest the following dosing recommendations for optimal glutamine benefits: (1) Total daily glutamine supplementation should be between 0.15 and

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0.30 g/kg/d (10.5–21 g for 70 kg individual), as these amounts increased glutamine levels without adverse effects [90, 91]; (2) For acute supplementation approaches, we recommend separating the amount into several smaller doses and ingesting before and after HS, e.g., consuming 0.075 g/kg/serving before and after HS increased and stabilized plasma glutamine levels [92]. A large single bolus should be avoided to prevent GI disturbance; and (3) Chronic supplementation strategies should follow similar guidelines with the daily dose split into multiple smaller quantities. Seven-day cycles or less appear to be optimal for raising glutamine levels prior to HS. The impact of longer supplementation routines is currently not known.

6. Conclusions

The forecasted occurrences of heat injury and death related to HS among those working in the heat is startling and agencies such as the U.S. Occupational Safety and Health Administration, International Labor Organization, and American Conference of Governmental Hygienists have raised alarms and published prevention strategies [33, 182]. The primary goal is to prevent a severe rise in core temperature through proper hydration and active cooling behaviors (e.g., work/rest cycles, appropriate clothing, cooling technologies). Functional nutritional aids offer another possible strategy to partner with prevention strategies. Glutamine is an ideal supplement to serve this function as it's been shown to address many of the mechanistic underpinnings of HS-related illness such as oxidative stress, GI damage, and inflammatory dysregulation. Benefits have been reported after supplementation with a mild dose of glutamine before and after HS. Continued research is needed in humans to better understand important inflammatory regulators during and after HS and the role of nutraceuticals, including glutamine.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Conflicts of Interest

The authors declare no conflicts of interest.

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